

Q.6c (15 marks): Describe the twin method to study the effects of heredity and environment upon the physical and...

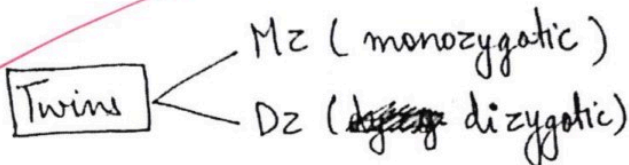
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(c) Describe the twin method of studying the effects of heredity and environment upon the physical and mental development of the individual

15

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Twins are the offsprings from same pregnancy.



Twin method uses the genetic background of the twins to study the effect of heredity and environment factors in development of certain traits.

1) Heritability: It is role of genetic variation against total variation among twins. It is calculated using concordance rates.

$$H = \frac{V_G}{V_G + V_E} = \frac{V_{DZ} - V_{MZ}}{V_{DZ}}$$

V_G = genetic variation

V_E = environment variation

V_{DZ}/V_{MZ} : monozygotic and dizygotic variation.

These with examples uses the shown ✓

If $H=1$, it is only totally due genetic factors, while if it is 0 then it is due to environment factor only.

- 2) ~~Calculation~~ Calculation of number of MZ and DZ in a population is done by Hardy-Weinberg ~~diff~~ differential methods.

Applications

- 1) Study of polygenic traits.
- 2) Effect of environment and genetic factors on variations.
- 3) NASA: effect of space environment on astronauts.
- 4) Study of obesity on ^{twin} ~~the~~ soldiers by US army.

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Limitation

1) It does not specify the exact genetic and environment component.

2) It does not specify the inheritance pattern of traits.

3) Only applicable on twins not whole population.

Hence, twin methods along with complementary methods like foster-child, co-twin method help anthropologists to learn about nature nurture aspects in development of trait.